Yingxuan You

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Education

École Polytechnique Fédérale de Lausanne (EPFL) *Lausanne, Switzerland*
PhD Student in Computer Science. Advisor: Prof. **Pascal Fua**. 2024.09 – Present
Research Topics: Physically Grounded 3D Generation, Geometric Computing, and 3D Vision.

Peking University (PKU) *Beijing, China*
Master Student in Computer Science. Advisor: Prof. **Hong Liu**. 2021.09 – 2024.07
Research Topics: 3D Human Pose and Shape Estimation. **GPA: 3.91 / 4.0**

Beihang University (BUAA) *Beijing, China*
Bachelor of Automation Science. 2017.09 – 2021.06
Bachelor Thesis: Multimodal Semantic SLAM for Dynamic Environments. **GPA: 3.84 / 4.0**

Publications

- **Spatio-Temporal Garment Reconstruction Using Diffusion Mapping via Pattern Coordinates**
  Yingxuan You*, Ren Li*, Corentin Dumery, Cong Cao, Hao Li, Pascal Fua. (* Equal Contribution)
  *IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**)*, 2026.
- **PhysGen: Physically Grounded 3D Shape Generation for Industrial Design**
  Yingxuan You, Chen Zhao, Hantao Zhang, Mingda Xu, Pascal Fua.
  *IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026. **Project Page**
- **Flow-Corrected Shape Optimization: Taming Manifold Drift in High-Dimensional 3D Models**
  Emilien Seiler, Nicolas Talabot, **Yingxuan You**, Federico Stella, Pascal Fua.
  Accepted by *The Conference on Neural Information Processing Systems (**NeurIPS**)*, 2026.
- **Global Graph-Validated Optimization for VLM-based 3D Indoor Scene Generation**
  Jialu Huang, **Yingxuan You**, Fei Wang, Zheng Dang.
  *The European Conference on Computer Vision (**ECCV**)*, 2026.
- **Single View Garment Reconstruction Using Diffusion Mapping Via Pattern Coordinates**
  Ren Li, Cong Cao, Corentin Dumery, **Yingxuan You**, Hao Li, Pascal Fua.
  SIGGRAPH, 2025. **Project Page**
- **Co-Evolution of Pose and Mesh for 3D Human Body Estimation from Video**
  Yingxuan You, Hong Liu, Ti Wang, Wenhao Li, Runwei Ding, Xia Li.
  *IEEE International Conference on Computer Vision (**ICCV**)*, 2023. **Project Page**
- **GATOR: Graph-Aware Transformer with Motion-Disentangled Regression for Human Mesh Recovery from a 2D Pose**
  Yingxuan You, Hong Liu, Xia Li, Wenhao Li, Ti Wang, Runwei Ding.
  *IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**)*, 2023.
- **ARTS: Semi-Analytical Regressor Using Disentangled Skeletal Representations for Human Mesh Recovery from Videos**
  Tao Tang, Hong Liu, **Yingxuan You**, Ti Wang, Wenhao Li.
  *ACM International Conference on Multimedia (**ACM MM**)*, 2024.
- **Dual-Branch Graph Transformer Network for 3D Human Mesh Reconstruction from Video**
  Tao Tang, Hong Liu, **Yingxuan You**, Ti Wang, Wenhao Li.

*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*, 2024.

- **An Efficient Graph Transformer Network for Video-Based Human Mesh Reconstruction**
  Tao Tang, **Yingxuan You**, Ti Wang, Hong Liu.
  *CAAI International Conference on Artificial Intelligence (**CICAI**)*, 2023.
- **MISD-SLAM: Multimodal Semantic SLAM for Dynamic Environments**
  Yingxuan You, Peng Wei, Jialun Cai, Weibo Huang, Risheng Kang, Hong Liu.
  Wireless Communications and Mobile Computing, 2022.

Work Experience

Alibaba, *Hangzhou, China* 2023.07 – 2023.12

Research Intern in Digital Human Team. Topic: 3D Face Reconstruction & Texture Generation.

- Explore effective constraints on face texture generation to fine-tune diffusion model w/o 3D scanned data.
- Propose DiffFace and self-supervised constraints to learn high-quality and renderable texture from image.
- Design a side branch with a condition input to beautify the original texture and maintain personal identity.

EasyHome, *Beijing, China* 2021.03 – 2021.08

Research Intern in SLAM Team. Topic: SLAM & 3D Scene Reconstruction.

- Study semantic SLAM for indoor scenes that combines the visual SLAM with the segmentation network.
- Alleviate the impact of dynamic pixels using semantic information and multi-view geometric constraints.
- Scan and build a housing dataset. Test algorithm effectiveness on IOS & Android in real-world practices.

Invention Patent

- **A 3D Human Mesh Reconstruction Method Based on Graph Skeleton Attention.**
  Hong Liu, **Yingxuan You**, Yang Chen, Wenhao Li.
  Invention patent, Published Application Number: CN115294265A, 2022.
- **A 3D Human Pose Estimation Method Based on Interweaved Graph and Attention Network.**
  Hong Liu, Ti Wang, Wenhao Li, **Yingxuan You**, Runwei Ding.
  Invention patent, Published Application Number: CN116129051A, 2023.

Awards and Honors

- EDIC fellowship, *EPFL* 2024
- Outstanding Graduate Award, *Peking University* (**Top 5%**) 2024
- Merit Student Scholarship, *Peking University* (**Top 5%**) 2022, 2023
- Outstanding Graduate Award, *Beihang University* (**Top 5%**) 2021
- First Prize of Segway-Ninebot Scholarship, *Beihang University* (**Top 1%**) 2021
- First Prize of Competition Scholarship, *Beihang University* (**Top 2%**) 2019, 2020

Skills

- **Programming**: Python, Pytorch, C/C++, MATLAB, LaTeX.
- **Languages**: Mandarin (Native), English (Fluent, IELTS: 7.0), French (A1).

Open Source

Codes for my published papers are available on my GitHub:

- (TPAMI 2026) DMap: https://github.com/kasvii/DMap
- (CVPR 2026) PhysGen: https://github.com/kasvii/PhysGen
- (ICCV 2023) PMCE: https://github.com/kasvii/PMCE
- (ICASSP 2023) GATOR: https://github.com/kasvii/GATOR